Age-Related Variation and Sex-Based Comparisons in Insular Cortical Thickness in Youth

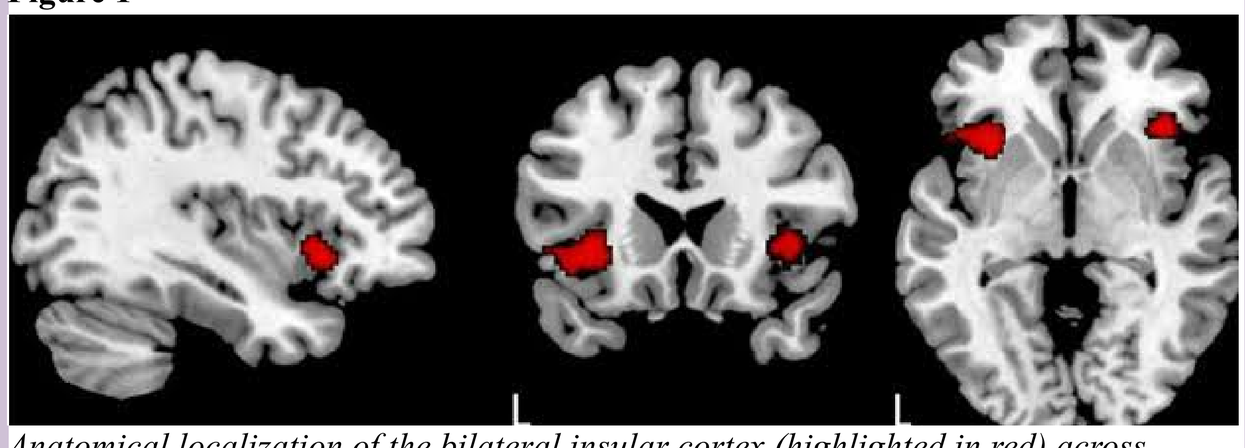
Charlotte Sumpf[1], Amy Jolin[1], Rachel Cluett[1], Jacob A. Burack[1], Tasmia Hai[1,2]

[1]McGill University; [2]University of Manitoba

Background

- **Structural & Functional Role:** The insular cortex is a structurally heterogeneous brain region involved in interoception, salience processing, and the integration of sensory, emotional, and bodily information. These functions support awareness of internal states and responses to environmental demands.
- **Developmental Maturation:** Childhood and adolescence are characterized by ongoing cortical maturation. Examining insular cortical thickness during this period can help characterize age-related structural variation in a region that supports multiple cognitive and socioemotional processes.
- **Individual Differences:** Brain maturation varies across individuals, brain regions, and potentially hemispheres and sex. Characterizing age-related and sex-based variation in bilateral insular thickness can help clarify patterns of structural variability in youth, providing a foundation for future studies of typical and atypical neurodevelopment.
- **Study Purpose:** To investigate associations between age and bilateral insular cortical thickness and examine sex-based differences in insular thickness among youth aged 8–16 years.

Figure 1



Anatomical localization of the bilateral insular cortex (highlighted in red) across sagittal, coronal, and axial MRI views.

Method

- **Participants:** N = 36 youth aged 8–16 years (16 male, 20 female) recruited from the Greater Montreal Area.
- **MRI Acquisition:** High-resolution T1-weighted structural images acquired using a MPRAGE sequence on a 3T Siemens Prisma Fit scanner with a 32-channel head coil (TR = 2300 ms, TE = 2.29 ms, TI = 900 ms, flip angle = 8°, voxel size = 0.8 mm³, FOV = 230 mm; ~5 min).
- **Image Processing:** MRI data extracted using CBRAIN and processed in FreeSurfer 7.4 for cortical reconstruction and parcellation. Mean cortical thickness was extracted separately for the left and right insula.
- **Statistical Analysis:** Separate linear regression models examined associations between age and bilateral insular cortical thickness. Sex (male/female) was included as a categorical predictor to assess sex-based differences in cortical thickness.

Characteristics	Total sample	Boys	Girls
Sample Size (N)	36	16	20
Age: Mean (SD)	12.28 (2.60)	11.88 (2.87)	12.60 (2.39)
Left insular thickness: Mean(SD)	3.10 (0.16)	3.09 (0.18)	3.10 (0.15)
Right insular thickness: Mean(SD)	3.15 (0.15)	3.19 (0.15)	3.12 (0.14)

Results

Region	Mean (SD)	Age B (SE)	β	P	95% CI	R^2	Sex B (p-val)
Right Insula	3.15 (0.15)	-0.026 (0.009)	-0.46	**.005**	[-0.043, -0.008]	26.1%	-0.054 (.229)
Left Insula	3.10 (0.16)	-0.028 (0.010)	-0.42	**.008**	[-0.048,-0.008]	19.3%	0.029 (.577)

- **Descriptive Patterns:** Mean cortical thickness was slightly greater in the right than the left insula. This descriptive difference provides context for the bilateral analyses but does not establish a statistically significant hemispheric difference.
- **Age-Related Variation:** Older age was associated with lower cortical thickness in both hemispheres, indicating a consistent pattern of insular thinning across the 8-16 year age range. Age explained approximately 19-26% of the variance in the respective regression models, with a stronger association observed for the right insula.
- **Sex-Based Differences:** No significant sex-based differences in age-adjusted cortical thickness were observed in either hemisphere. Within this sample, the results therefore provide no evidence of differences in mean insular thickness between male and female youth after accounting for age.

Figure 1. Age-Related Variation:
Insular cortical thickness across childhood and adolescence

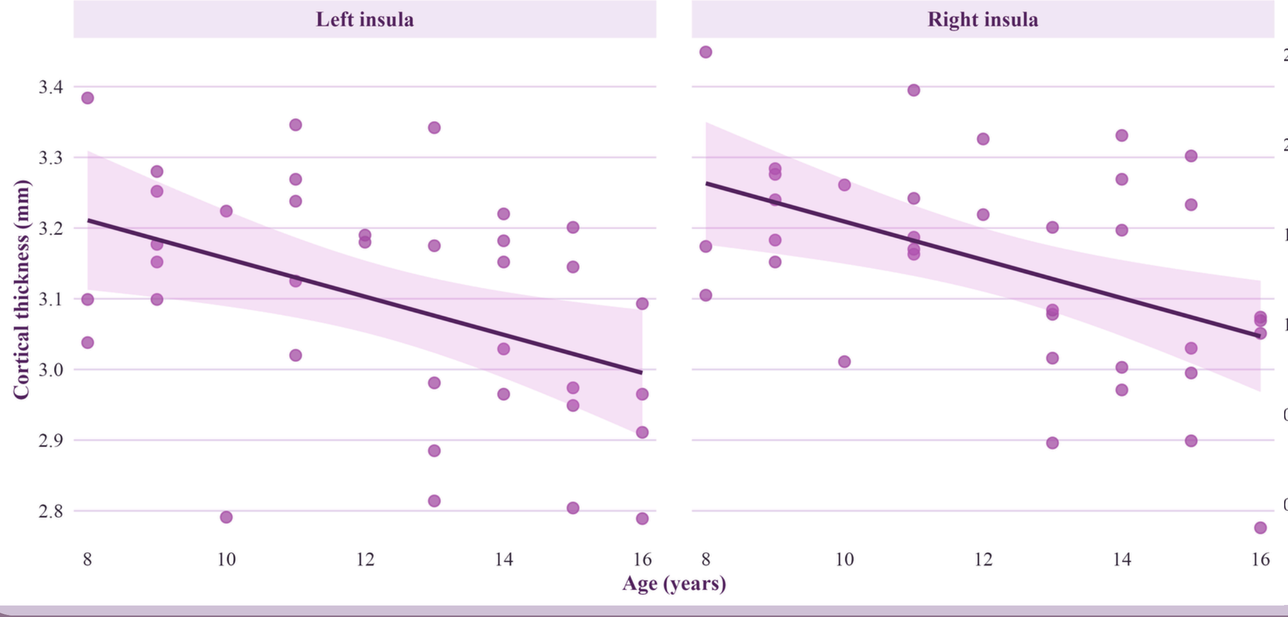
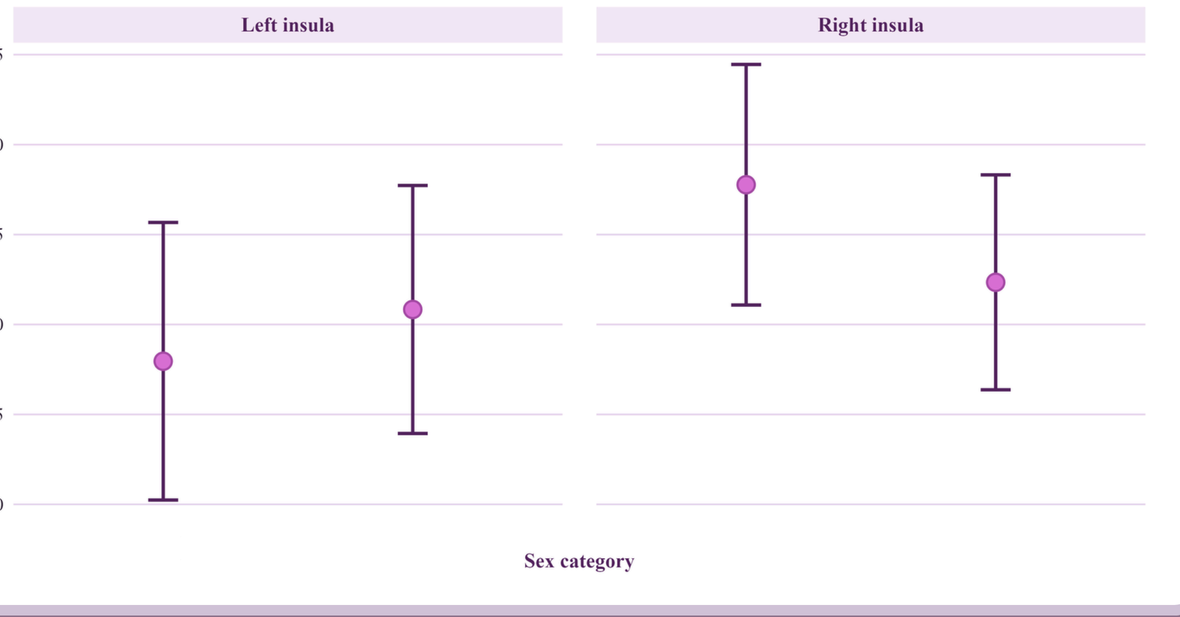


Figure 2. Sex-Based Comparison:
Insular cortical thickness across sex



Discussion

- **Age-Related Variation:** Older age was associated with lower cortical thickness in both the left and right insula, suggesting age-related variation in insular structure across youth aged 8–16 years.
- **Hemispheric Patterns:** Similar age-related associations were observed across hemispheres, providing preliminary evidence of comparable patterns of age-related variation in bilateral insular thickness.
- **Sex-Based Differences:** No statistically significant sex-based differences in age-adjusted insular cortical thickness were detected.

Limitations & Future Research

- **Sample Size:** The small sample (N = 36) limits statistical power, precision, and generalizability. Larger, more diverse samples are needed to evaluate the robustness of the observed associations and better characterize individual variability.
- **Developmental Trajectories:** The cross-sectional design captures age-related differences at one time point rather than within-person change. Longitudinal studies could clarify how insular cortical thickness changes throughout childhood and adolescence.
- **Future Directions:** Future research could examine potential hemispheric and sex-related differences in larger samples and integrate cortical thickness measures with diffusion MRI and tractography to investigate relationships between insular anatomy and structural connectivity.

Scan for references

This research was supported in part by funding from the Canada First Research Excellence Fund, awarded through the Healthy Brains, Healthy Lives initiative at McGill University.

Contact Us
Charlotte.Sumpf
@mail.mcgill.ca